

Drones in Combat: An Analysis with a Focus on Ukraine

778 Words

Introduction:

Unmanned Aerial Vehicles (UAVs), more commonly known as drones, have emerged as transformative tools in modern warfare, offering enhanced surveillance, reconnaissance, and strike capabilities. The conflict in Ukraine provides a compelling case study of the utilization of drones in combat, generating instances of the upsides and caveats of applying these technologies on the battlefield. In short, this paper is meant to elaborate on the effectiveness of these unmanned systems derived from use cases on the ground, as well as the countermeasures developed to neutralize them and understand the significant implications of these vehicles on the future of warfare.

Utilization of drones in Ukraine:

The ongoing conflict in the Donbas region has witnessed a significant deployment of drones by both parties. Both Ukrainian forces and separatist groups have leveraged UAVs for intelligence gathering, target acquisition, and even precision strikes. Drones have been instrumental in monitoring troop movements, identifying enemy positions, and providing real-time situational awareness on the battlefield^[1].

One notable example is the use of Turkish-made Bayraktar TB2 drones by the Ukrainian Armed Forces. These UAVs have been employed for surveillance and reconnaissance missions, offering an efficient solution for monitoring the dynamic and complex battlefield^[2]. The ability of drones to operate in contested airspace, providing valuable information without risking human lives, has undoubtedly altered the nature of conflict in the region.

Efficacy of drones:

Drone effectiveness in the Ukrainian conflict is evident in their contribution to tactical successes on the battlefield. Real-time information provided by UAVs has enabled Ukrainian forces to respond rapidly to enemy movements and adjust their strategies accordingly. The precision strike capabilities of armed drones have been particularly impactful, allowing

Ukrainian forces to target key assets of separatist groups with reduced collateral damage^[3].

Drone usage has also increased mobility for ground forces with access to lower-level equipment while delivering most of the same functionality required of more complex aerial vehicles, i.e., overwhelming anti-aircraft defenses, deploying explosives, and gaining intelligence. For instance, China's DJI Mavic 3, used by both Ukrainian and Russian forces, costs around USD 2000, a stark contrast from the USD 110 million (starting price) required for an F-35^[4]. While a fighter plane of that magnitude enables its users

However, the use of drones in combat has not been without challenges. Adverse weather conditions, electronic warfare, and evolving countermeasures have posed threats to drone operations. As drones become more integral to military strategies, addressing these challenges becomes imperative for sustaining their effectiveness on the battlefield.

Countermeasures against drones:

The proliferation of drones has prompted the development of countermeasures to neutralize or mitigate their impact. Anti-drone technologies, including electronic warfare systems, signal jamming devices, and dedicated anti-drone weapons, have been deployed to disrupt drone operations^[5]. Furthermore, the development of effective detection systems has become crucial in identifying and tracking UAVs, allowing military forces to take proactive measures.

Russian-backed separatist forces in Ukraine, for instance, have employed electronic warfare systems to disrupt the communication links between Ukrainian forces and their drones^[6], creating a cat-and-mouse game between drone operators and those seeking to neutralize them highlights the evolving nature of modern warfare.

For cheaper equipment, the problems are more noticeable, as logistics problems plague Ukraine and its surrounding regions, soldiers are forced to either resort to backdoors in the

industry for a consistent supply of cheaper drones or utilize more expensive measures guaranteed by Western diplomacy to keep the war favorable.

Conclusion:

The use of drones in the conflict in Ukraine underscores the transformative impact of unmanned systems on contemporary warfare. Their ability to provide real-time intelligence and precision strikes has proven effective on the battlefield. However, the emergence of countermeasures and challenges such as electronic warfare highlights the need for ongoing technological innovation and adaptation.

As the technology behind drones continues to evolve, their role in conflict zones will likely expand, necessitating a comprehensive understanding of both their capabilities and vulnerabilities. The Ukrainian conflict serves as a valuable case study for military strategists and policymakers seeking to navigate the complex terrain of modern warfare, where the dynamics between unmanned systems and countermeasures will shape the future landscape of armed conflict.

Citations:

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